The Bedrock of Economic Geography

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Abstract

Why do economic maps persist after the advantages that first supported settlement have weakened? I examine the agricultural foundations of a persistent regional geography in Skåne, southern Sweden, linking bedrock maps to several centuries of agricultural and population records and to modern workplace data. Areas with more carbonate-rich bedrock had greater agricultural capacity and denser populations throughout the historical record. The historical population map thus reflects variation in the region's agricultural capacity. Moreover, population records from centuries before industrialization already predict where people live today. The early nineteenth-century population map also predicts where workplaces, employment, and value added are concentrated today. Historically, the relationships of bedrock with agricultural capacity and population density extend beyond the areas surrounding major cities. Today, the relationship between bedrock and residential density is concentrated around these centers. This pattern is consistent with agricultural districts supporting towns and villages whose markets, buildings, and connections remained valuable as the economy changed. Historically populous places remain important locations for both residence and production, showing the lasting economic importance of the region's agricultural settlement geography.

Keywords: Economic geography; historical persistence; agricultural capacity; settlement.
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1 Introduction

Why do economic maps persist after the original productive advantage has faded? Even inside modern high-income countries with common formal institutions, people, workplaces,

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employment, and production are sharply concentrated. Contemporary concentration is sustained by market access, infrastructure, labor pooling, input-output linkages, and agglomeration economies. These forces do not rule out deep historical origins. They can be the channels through which an older economic map, formed when agricultural surplus and settlement depended more directly on the physical landscape, remains economically relevant after the initial advantage has weakened (Krugman, 1991; Ellison and Glaeser, 1999; Head and Mayer, 2004, 2006; Davis and Weinstein, 2002). I study one of the deepest such fundamentals: bedrock.

Bedrock and surface deposits help shape soil properties. For example, carbonate-rich rocks, such as limestone and marl, contain minerals that can neutralize soil acidity and improve growing conditions where acidity limits crop production. Together with drainage and topography, these properties influence agricultural capacity and the surplus available to support settlement. These agricultural conditions can in turn shape where villages, churches, roads, and market centers develop. Once a settlement system exists, its usefulness extends beyond the fields that first supported it. Roads connect established places; buildings embody earlier investment; firms gain access to workers and customers. An advantage that first mattered through agriculture can acquire a new economic life through the places built around it.

This paper traces the agricultural foundations of a persistent economic geography in Skåne, southern Sweden. Linking bedrock maps to historical agricultural and population records and to modern economic activity reveals the agrarian economy behind the inherited settlement pattern. Areas with more carbonate-rich bedrock had greater agricultural capacity and denser populations long before industrialization. This historical population geography still predicts where people live and work, although the modern residential relationship with bedrock is concentrated around major cities.

Skåne offers considerable geological variation within a small, integrated regional economy. Carbonate-rich formations lie alongside crystalline and other sedimentary rocks, and rich historical sources record agricultural assessments, population, churches, and towns. The region belonged to Denmark until 1658, when it became part of Sweden. The same landscape can therefore be followed across two systems of state administration and historical recordkeeping.

The geological measure is the carbonate-rich share of each historical parish or city area. The main historical outcomes are population density and *mantal* per square kilometer. Mantal records the assessed taxable capacity of agricultural holdings, providing evidence on the agricultural base of the local economy alongside its population. Using the same historical boundaries, I measure modern population density in 2025 and mean annual densities of

establishments, employment, and value added from private-sector workplace records over 1997–2021.

Carbonate-rich bedrock is strongly and consistently related to historical agricultural capacity and population density. The estimates are positive for both outcomes in all five historical years from 1570 to 1810, conditional on location and area size. The pooled coefficients remain statistically significant at the 1 percent level with these controls. The population coefficient is almost identical at the beginning and end of the period, despite Skåne's transfer from Danish to Swedish rule. The correspondence between assessed agricultural capacity and population links early settlement concentrations to the resources available to support farming households and demand for trade and services.

The settlement evidence complements these parish-level comparisons. Churches, towns, and the broader settlement records are more concentrated in areas with larger carbonate-rich shares. Across the ten historical town areas in 1810, larger populations are associated with greater total agricultural capacity in the wider surrounding countryside. This relationship becomes stronger as the hinterland is extended from 10 to 30 kilometers. Together, these findings are consistent with agricultural districts supporting both rural communities and centers serving a wider market.

Population records then connect this agrarian economy to the modern distribution of activity. I compare population density in 1570, 1750, and 1810 as predictors of density in 2025, using the same 381 areas and controls. The 1810 map provides the best single-year fit, but much of its explanatory contribution is already present in 1570. Moving from the twenty-fifth to the seventy-fifth percentile of population density in either 1570 or 1810 corresponds to about 0.44 standard deviations of transformed modern population density, conditional on bedrock, location, and area. The persistent pattern is thus quantitatively substantial and reaches back centuries before Sweden's broad industrial transformation.

Including population density in 1810 reduces the modern residential coefficient on bedrock by almost half. This decline links the modern geological relationship to a population geography already established in the agrarian economy. Historical population also strongly predicts the densities of establishments, employment, and value added over 1997–2021, conditional on geology and the geographical controls. An inherited housing stock or local amenities could sustain residential concentrations while people commute elsewhere for work. Persistence in employment and production shows that the historical geography extends to where economic activity takes place.

The relationship with bedrock nevertheless differs in its geographical reach across periods. Historical population and agricultural capacity remain positively related to carbonate-rich share outside the areas surrounding major cities. The modern residential coefficient, by

contrast, is close to zero when areas within 10 kilometers of the major city centers are excluded. This contrast is consistent with agricultural districts retaining their importance through established centers as employment shifted toward industry and services. Markets, buildings, transport connections, and labor pools offer possible sources of that continuity.

The paper contributes to research on natural advantage and agglomeration (Krugman, 1991; Duranton and Puga, 2004; Rosenthal and Strange, 2004; Puga, 2010; Ellison et al., 2010; Redding and Rossi-Hansberg, 2017). Combes et al. (2010) show that geology and historical urban population predict modern employment density, using this variation to estimate agglomeration effects on wages and productivity. Their geological measures include limestone and marl, and they motivate the instruments partly through agriculture's influence on settlement. I examine this link through joint records of agricultural capacity and population across rural parishes and towns. The successive observations place this relationship centuries before industrialization and show its continuity under Danish and Swedish rule. The contribution is to establish the agrarian economic content and historical depth of the population geography inherited by the modern economy.

The agricultural evidence also relates to research on preindustrial population and city formation. Ashraf and Galor (2011) document a positive relationship between land productivity and preindustrial population density. Bosker and Buringh (2017) show how transport geography and proximity to existing cities shaped European city formation. In a dynamic model grounded in historical U.S. evidence, Nagy (2023) explains how agricultural hinterlands support city formation and growth. The Scanian records bring agricultural capacity and population into the same comparison, within parishes and between towns and their surrounding countryside.

Structural transformation provides a link between agricultural geography and modern concentration. Henderson et al. (2018) find that agricultural geography has greater explanatory importance relative to trade geography in countries that developed earlier, even though agriculture now employs a smaller share of their workers. They relate this pattern to early agglomerations persisting as food became cheaper to transport. Michaels et al. (2012) show how the shift out of agriculture increased population concentration across rural and urban locations in the United States, with growth depending on initial density and the agricultural employment share. The contrast in Skåne between a broad historical agricultural relationship and a modern residential relationship concentrated around cities is consistent with this reorganization of activity within an inherited settlement system.

A further literature examines persistence through major changes in locational advantages. Davis and Weinstein (2002) document the recovery of Japanese cities after wartime destruction. Bleakley and Lin (2012) find persistent population concentrations at obsolete

American portage sites, while Jedwab et al. (2017) show that Kenyan railway cities survived railway decline and the departure of settlers and traders. In Skåne, agriculture's economic role changed gradually: bedrock remained fixed and farming continued, while established places attracted activity under a different employment structure. The agricultural assessments help explain the economic basis of those early population concentrations.

Persistence also depends on adaptation. Michaels and Rauch (2018) show that Britain's urban network adjusted more readily to new transport needs after Roman rule than France's; German division and reunification demonstrate the importance of changing market access (Redding and Sturm, 2008; Ahlfeldt et al., 2015). Industrial policy yields contrasting outcomes: manufacturing employment gains in the Tennessee Valley persisted after federal transfers ended (Kline and Moretti, 2014), whereas Chinese counties hosting large industrial plants experienced early productivity gains followed by relative decline (Heblich et al., 2026). These studies distinguish the persistence of established concentrations from their consequences for subsequent growth. Allen and Donaldson (2020, 2022) formalize the roles of contemporaneous and historical spillovers in spatial persistence.

The paper also relates to the broader literature on geography, early technology, and population history in modern development (Olsson and Hibbs, 2005; Nunn, 2009; Comin et al., 2010; Spolaore and Wacziarg, 2013), following these relationships at a local scale within a common modern institutional setting.

2 Historical Setting and Conceptual Framework

2.1 The Scanian Landscape

Skåne lies at the southern end of Sweden, facing Denmark across the Öresund (Figure 1). Several major bedrock environments occur within this small region: carbonate-rich formations lie alongside other sedimentary rocks, mafic and intermediate rocks, and crystalline landscapes. The region includes plains, intermediate farming districts, and forested areas with different cultivation conditions and settlement histories. The comparison thus follows variations in the agricultural landscape within a regional economy whose residents and firms share national institutions today.

Bedrock matters for agriculture through the conditions under which land is cultivated. Carbonate-rich rocks and sediments affect soil chemistry, including acidity and nutrient availability. Their significance for farming also depends on surface deposits, drainage, terrain, and cultivation practices (Jenny, 1941; Schaetzl and Anderson, 2005; Food and Agriculture Organization of the United Nations, 1973). The bedrock measure captures broad geological

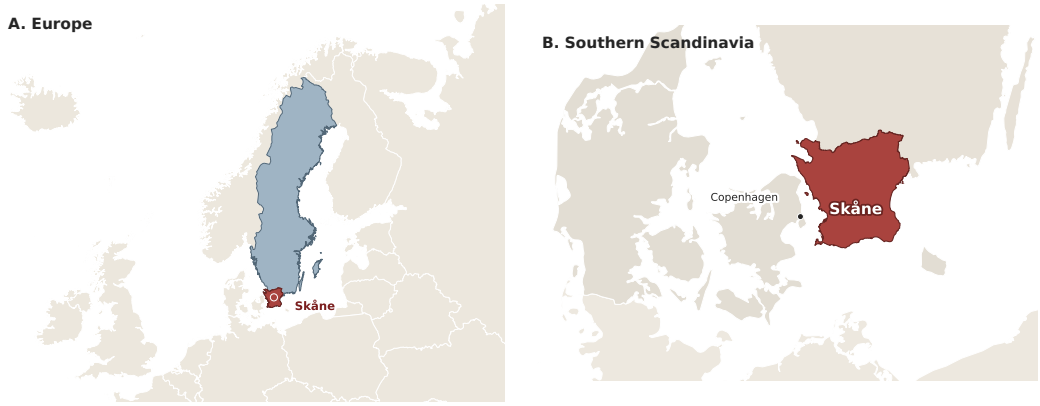


Figure 1: Skåne in northern Europe
Sources: Natural Earth (country boundaries); GADM 4.1 (regional boundary).

differences relevant to agriculture. Within each area, soil quality and cultivation conditions can however still vary considerably.

Cultivated soils need not have formed from the bedrock directly beneath them. Surface deposits can modify the connection between bedrock and soil chemistry; drainage and terrain can make chemically favorable land difficult to farm. The empirical question is whether these geological differences nevertheless correspond to the agricultural economy recorded in the historical sources. Agricultural assessments and population records allow this relationship to be followed through centuries of cultivation and adaptation.

Work on Scanian agriculture describes grain production, crop and animal husbandry, and the organization of cultivation within villages (Skoglund, 2022; Lageråas and Magnell, 2020). Farming took place within systems of landholding and local institutions that organized the use of land and the distribution of its produce. Agricultural capacity could thus be reflected in the number of households an area supported, the fiscal assessments of its holdings, and the trade and services sustained in nearby settlements. These indicators describe complementary aspects of the local economy.

The historical records also straddle a major change in state authority. Skåne belonged to Denmark until the Treaty of Roskilde in 1658, when it became part of Sweden. The early fiscal and population observations can consequently be compared across Danish and Swedish source traditions. A geographical gradient visible through this transition is evidence of continuity in the region's agrarian economy. Differences in assessment practices and source coverage can affect comparisons across periods, particularly for mantal.

2.2 Agricultural Capacity and Persistent Settlement

An area with greater agricultural capacity can support more farming households and supply nearby settlements with produce. These households also generate demand for trade, crafts, and services. Specialized trades require enough customers to cover the costs of operating, and costly travel limits the market available to any one location. A larger agricultural population can therefore sustain a wider range of nonfarm activities within reach of its farms. Gathering these activities in a common center allows households to obtain several goods and services in one visit and gives traders access to buyers and suppliers. Repeated use of the same places encourages investment in market facilities, buildings, and connecting roads. Towns and villages can consequently draw their importance from the agricultural hinterlands they serve. Nagy (2023) formalizes this connection in a model where access to agricultural hinterlands raises the returns to nonfarm production and supports city formation and growth. The agricultural base of a town can therefore extend well beyond its own boundaries. Town and village locations need not coincide with the most productive land: access to surrounding households and routes of exchange also matters. Agricultural conditions can thus shape both rural population density and the network of local centers.

Historical records of agricultural capacity and population provide complementary evidence on this regional economy. Mantal records the assessed agricultural capacity of local holdings, while the population figures encompass both farming households and those supported by trade and services. A shared association with bedrock links the historical population pattern to the agricultural resources that could sustain it. Landholding and taxation influence the distribution of those resources, while migration and family formation affect where people live. The outcomes studied here concern capacity and people per unit of land. Favorable agricultural conditions could sustain a larger population at similar living standards.

2.3 Why Established Places Can Remain Valuable

Once a settlement has formed, accumulated investments change the relative attractiveness of locations. Buildings and transport links are costly to reproduce and can remain useful to successive activities. A firm entering an established town may find premises, workers, suppliers, and access to customers already in place. At a new location, some of these advantages would have to be assembled before production could begin. Commercial relationships and familiarity with a market can also retain value as the goods being traded change. These inherited advantages give firms and households a reason to continue using an established location after agriculture has become a smaller part of its economy. Current agglomeration

economies can reinforce that advantage through labor-market matching, specialized suppliers, and access to customers (Duranton and Puga, 2004; Rosenthal and Strange, 2004; Puga, 2010). Allen and Donaldson (2020, 2022) develop this distinction between contemporaneous and intertemporal forces in spatial persistence.

Structural transformation can change the scale at which this inheritance remains visible. Farming requires workers to be close to cultivated land, while many industrial and service activities benefit from proximity to other firms and workers. If new employment concentrates in established centers, fewer households may depend on living near the fields, even as the district's towns attract people and production. An agricultural landscape that initially supported dispersed rural settlement can then leave a legacy concentrated in a smaller number of places. This outcome depends on the continuing usefulness of those centers. New transport routes or technologies may favor other locations and weaken the advantage of the inherited network.

2.4 Implications for the Empirical Comparisons

The agricultural foundation has a joint implication for the historical outcomes. Where carbonate-rich bedrock improves cultivation conditions, it should be associated with both greater agricultural capacity and higher population density. A population gradient without a corresponding capacity gradient would give weaker support to an agricultural explanation of settlement. The regressions therefore examine both outcomes after controlling for broad location and area. Repetition across dates shows whether the relationship extends across changes in population, cultivation, and state authority.

The formation of local centers gives a second implication. Agricultural districts with greater capacity can support more people engaged in trade, crafts, and other specialized activities. Town populations should therefore be positively associated with agricultural capacity in the surrounding countryside. A center's relationship with bedrock inside its own boundaries may be weaker, since its suppliers and customers can live farther away. The empirical comparisons examine both the broader association of settlement records with bedrock and the relationship between town size and rural capacity at several distances.

The third implication concerns the continuing economic value of established places. If their inherited assets and connections remain useful, historical population density should predict both modern residence and workplace activity, conditional on geology and the geographical controls. Persistence confined to residence could reflect housing or amenities; corresponding workplace relationships would also connect the old map to the location of

production. Establishments, employment, and value added capture different margins: employment depends on both establishment numbers and size, while value added also reflects industry composition and production per worker. Comparisons using several historical dates establish how far back this relationship can be observed. Differences in fit can reflect subsequent development as well as the quality of the historical records.

The fourth implication is geographical and depends on where nonfarm activity expands. If it concentrates in established centers serving agricultural hinterlands, the historical association of bedrock with capacity and population can extend across rural areas even as its modern association becomes concentrated around those centers. The city comparisons examine this change in geographical reach by estimating historical and modern relationships outside the same city neighborhoods. A contrast between the two periods would be consistent with the legacy of agricultural districts surviving through their central places.

3 Data

I combine geological maps with historical population and land-tax records, settlement registers, and modern data on residents and workplaces. All measures refer to historical parish and city areas. Using the same boundaries throughout allows population and economic activity in each area to be compared over time. Appendix Table A1 summarizes the sources and the construction of the measures.

3.1 Geology and Historical Parishes

The Geological Survey of Sweden (SGU) supplies the bedrock maps. Using SGU's classification of rock types, I group areas mapped as limestone or marl into a carbonate-rich bedrock category (Geological Survey of Sweden, 2024). Combining these maps with the historical boundaries gives the share of each parish or city underlain by these rocks. The other categories include mafic and intermediate rocks, other sedimentary rocks, and granite and gneiss. Appendix Figure A1 shows the geological map.

Measuring geology as a share captures differences between areas with the same dominant rock type but different proportions of carbonate-rich bedrock. In the 381 areas used in the regressions, the mean share is 0.363, and the 25th and 75th percentiles are zero and 0.980. Much of the variation is therefore between places with little carbonate-rich bedrock and places where it underlies almost the entire area.

Historical observations come from the Sweden 1570–1810 parish panel, available through the Swedish National Data Service (Andersson Palm, 2025). It brings together information

on population, agriculture, land ownership, and tax assessments around 1570, 1630, 1690, 1750, and 1810. Even the final observation, in 1810, predates Sweden's broad industrial transformation in the second half of the nineteenth century (Ericsson and Molinder, 2020, p. 817). I match the historical records to parish and city boundaries from Lantmäteriet using place names and identifiers. Where several records refer to the same area, I sum their population or mantal values before calculating density.

The panel records population in 381 parish and city units at five historical dates, yielding 1,905 observations. Mantal is observed for 1,894 of these area-year observations. All 381 areas also have modern data on population density and employed-resident density. The comparisons using different historical dates therefore follow exactly the same places.

Cities are kept separate from neighboring rural parishes because their population and agricultural capacity can differ sharply. Combining them would obscure the distinction between a center and its surrounding countryside. Economic connections extend across these boundaries: residents of a rural parish may use markets, services, or workplaces in a nearby city.

Mantal measures the assessed taxable capacity of agricultural holdings. It reflects land quality, the extent and organization of cultivation, and the assessment practices of the authorities. I sum mantal within each historical area and divide by its area in square kilometers to measure agricultural-capacity density. Population density and mantal density are the main historical outcomes.

Mantal's fiscal origin matters for its interpretation. Assessments can remain unchanged as harvests fluctuate, and differences in taxation practices can affect comparisons across places. The population records provide complementary evidence on how many people lived in the same areas. Together, the mantal and population records connect the historical settlement pattern to the agricultural capacity of the land. The county comparisons in Section 7 examine sensitivity to broad administrative differences.

The relationship between assessed capacity and population can also change over time. Assessments may respond slowly to changes in cultivation, while migration and household formation alter population. I therefore examine the relationship of each outcome with bedrock at each historical date.

All densities are calculated using the full area of the historical parish or city. This common denominator allows agricultural capacity to be compared with the concentration of population and modern activity. Mantal density therefore depends on both the extent of agriculture within an area and the assessed capacity of its holdings.

For example, two equally large rural parishes may differ in mantal density because one contains more farmland, because its holdings receive higher assessments, or both. The measure captures the agricultural capacity concentrated within the parish, including variation in how much land is devoted to farming.

3.2 Settlement Records

Table 4 uses two measures of recorded settlement features per square kilometer. The first combines religious-building records from Bebyggelseregistret with an indicator for each of ten historical towns. A building enters when its earliest recorded historical event or building phase begins in 1810 or earlier. In the 381 areas used in the analysis, this gives 341 church records, two bell-tower records, and ten town indicators. A church and a town at the same location each contribute one entry. Together, these records locate early religious and urban centers.

The building dates describe the history recorded for a building or site, not necessarily the foundation of the surrounding settlement. Town indicators identify the historical city areas listed in Appendix A.1; Skanör and Falsterbo form one combined area.

The broader measure adds 20,123 mapped heritage features from the Swedish National Heritage Board's Fornsök register and 5,129 selected names from Lantmäteriet's Ortnamn database, giving 25,605 entries in total. The heritage features include occupation sites, village and farm plots, house foundations, hearths, cooking pits, church and monastery remains, mills, old routes, and rock carvings with cup marks. Selection uses the register's site-type and feature descriptions. The name selection includes church names and building or settlement names with forms associated with villages, farms, and estates. These additional features and names have no common pre-1810 date restriction.

For each measure, I count entries within the historical parish or city and divide by its full area in square kilometers. Each entry has equal weight, irrespective of its size. A heritage record can contain several mapped features, and different sources can describe the same place, so these are densities of recorded features and names rather than numbers of distinct settlements. Appendix A.1 gives the source categories and selection rules. The measures describe where religious and urban centers and a wider range of traces of habitation and local activity are concentrated.

For a separate comparison of towns and their agricultural hinterlands, I use the ten historical town areas with population observations in 1810. I locate their cores at central church sites documented before 1810, using coordinates from Bebyggelseregistret. Around each core, I sum rural mantal within 10, 20, and 30 kilometers, allocating each rural parish's

total in proportion to the share of its area within the circle. All city areas are excluded. Skanör and Falsterbo have a combined population observation, so I combine the circles around their two church sites, counting overlapping land once. The measure covers rural parishes within the Skåne study region and assumes uniform capacity within each parish; small areas without mantal observations are omitted.

3.3 Modern Residents and Workplaces

Statistics Sweden's small-area statistics provide population in 2025 and employed residents aged 20–64 in 2024 (Statistics Sweden, 2026a, 2025b). The data cover 819 demographic statistical areas, known as DeSO, using the 2025 boundaries for Skåne (Statistics Sweden, 2025a). I allocate their population and employed-resident counts to the historical areas in proportion to the geographical overlap, then divide by the full area of each historical parish or city.[1] For the maps in Figure 2, I use population in 2021 to match the final year of the workplace records, allocating it from Skåne's 789 DeSO areas on the 2018 boundaries using the same method (Statistics Sweden, 2026a).

For example, if a historical parish covers one quarter of a DeSO area, it receives one quarter of its reported population and employed residents. This allocation assumes that residents are evenly distributed within each DeSO area. Where settlement occupies only part of that area, the procedure can assign people to the wrong side of a historical boundary. The small size of the statistical areas limits the scale of this approximation, although it cannot reproduce the exact residential distribution within them.

Employed-resident density measures where working people live. The workplace records described below measure where employment is located. These can be different places: a person may live in a rural parish and commute to a nearby city. Keeping the measures separate allows the historical population map to be compared with both the residential and workplace geography of the modern economy.

Statistics Sweden provides restricted-access plant-level private-sector workplace records covering 1997–2021 (Statistics Sweden, 2026b). I use workplace counts, reported employment, and reported value added to measure economic activity per square kilometer. Recorded coordinates locate each workplace within a one-kilometer grid cell; each cell is assigned to the historical parish or city covering the largest part of its area. This approximation can affect assignments close to parish boundaries. The main results cover 1997–2021; the appendix also reports results for 2012–2021 and 2021, together with additional outcomes.

[1] Statistics Sweden makes small adjustments to the published population counts to protect confidentiality. Published components and totals can therefore differ slightly.

I first sum workplace counts, employment, and value added within each historical area for each year and divide by its area in square kilometers. I then average these annual densities over the years represented in each period. The regression uses the transformed average, giving one observation per parish or city. The full-period average describes the geography of recorded private activity over several decades; the shorter-period and single-year results show the pattern closer to the end of the available data.

Workplace density records the number of establishments per square kilometer. Employment density also reflects how many people they employ. Value-added density measures the concentration of production and depends on employment, production per worker, and industry composition. I use value added as reported for each workplace and year.

Figure 2 maps bedrock, population density in 1810 and 2021, and private workplace density in 2021 within the same historical boundaries. In 1810, population was relatively dense across southern and western Skåne, while the northern interior was more sparsely settled. Carbonate-rich bedrock is widespread in the south, although dense settlement also occurs on other formations along the western coast. The modern population map retains much of this broad geographical contrast, alongside local changes in relative density. Concentrations along the western coast and around several inland centers are more pronounced. The workplace map shows a similar broad pattern, with concentrations along the western coast and in the southwest. These panels connect the historical population geography to both modern residence and the location of private establishments. The regressions examine these relationships after accounting for location and the size of each area.

4 Empirical Design

I first examine whether areas with a larger share of carbonate-rich bedrock had greater agricultural capacity and denser populations. I then relate the settlement records to geology and compare town populations with agricultural capacity in the surrounding countryside. Finally, I examine how well the historical population maps predict the location of modern residents and workplaces. These comparisons follow the implications developed in Section 2.4, using the same historical boundaries throughout.

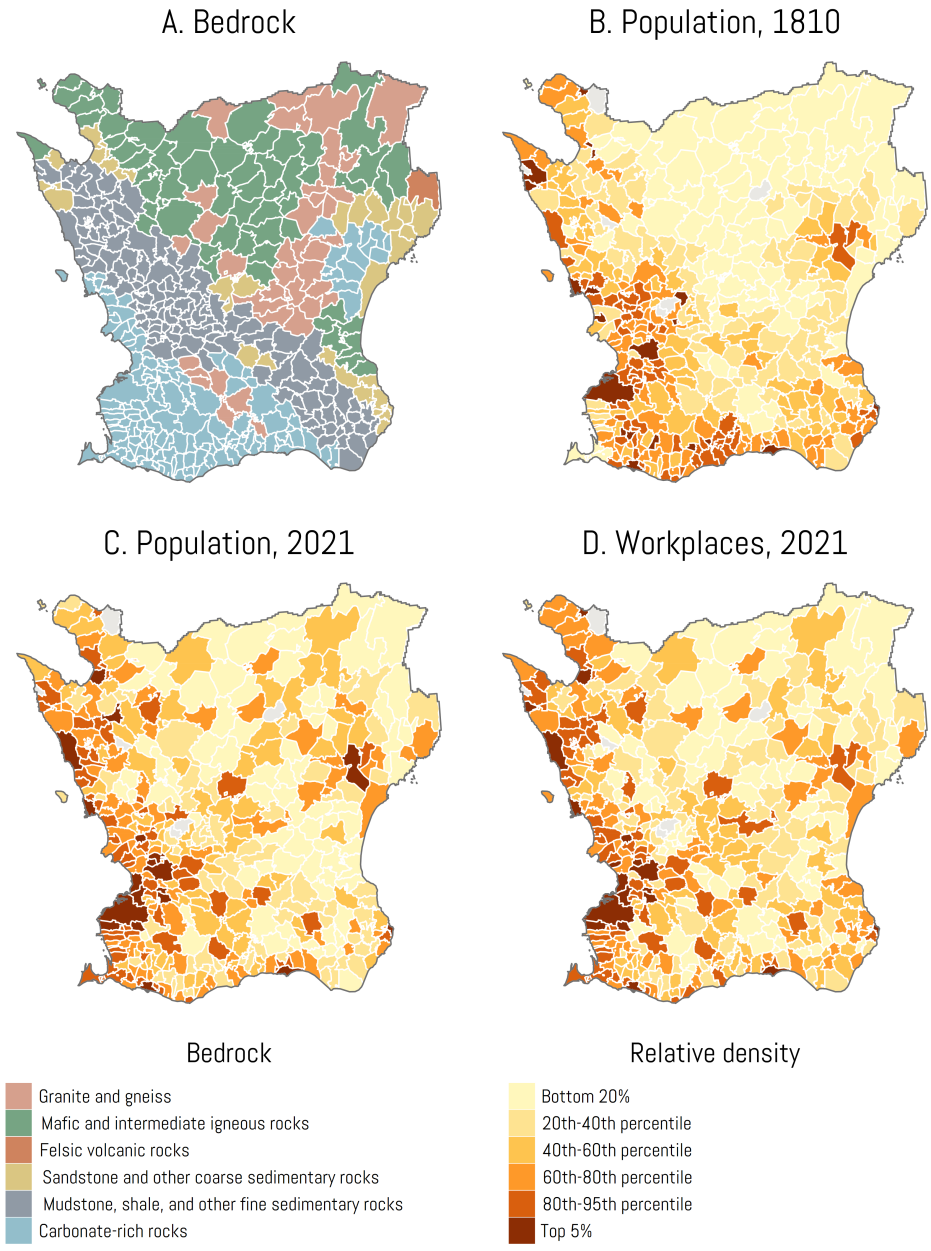


Figure 2: Bedrock and the geography of population and workplaces

Notes: Panel A shows the dominant bedrock group in each historical area. Panels B and C show population density in 1810 and 2021; Panel D shows private workplace density in 2021. Densities use the full historical area in square kilometers. Panels B–D rank the same 381 areas, with equal weight per area and separate 20th, 40th, 60th, 80th, and 95th percentile cutoffs. The shared legend shows relative position within each panel, not equal absolute density. Gray denotes areas not shown. Sources: SGU (bedrock), Sweden 1570–1810 through SND (historical population), Statistics Sweden (modern population and workplaces), and Lantmäteriet (historical boundaries). The coastline follows GADM 4.1.

Table 1: Summary statistics

Variable	Mean	SD	25th pct.	75th pct.	N
historical parish-year panel					
Historical population density	3.436	0.595	3.091	3.806	1,905
Historical mantal density	0.903	0.440	0.569	1.188	1,894
matched parish cross-section					
Carbonate-rich bedrock share	0.363	0.447	0.000	0.980	381
Population density, 1570	3.088	0.589	2.741	3.491	381
Population density, 1750	3.603	0.454	3.389	3.876	381
Population density, 1810	3.973	0.461	3.698	4.248	381
Modern population density	4.450	1.338	3.455	5.349	381
Employed-resident density	3.650	1.329	2.674	4.566	381
Church and town record density	0.053	0.053	0.016	0.074	381
Broader settlement record density	1.566	0.473	1.234	1.879	381

Notes: Density variables are reported on the transformed scale used in the regressions, with raw densities measured per square kilometer. The historical panel covers 1570, 1630, 1690, 1750, and 1810; mantal is missing for 11 parish-year observations. The matched cross-section contains 381 parish/city units at every historical year and for modern population and employed residents. Modern population refers to 2025 and employed residents aged 20–64 to 2024. Settlement measures count the entries defined in Section 3.2 per square kilometer; entries need not identify distinct settlements. Carbonate-rich bedrock remains on its original share scale.

4.1 Historical Agricultural and Population Density

Let G_i denote the carbonate-rich bedrock share of historical area i, and let q_{it} be population or mantal per square kilometer at date t. I estimate

$$q_{it} = \alpha + \beta G_i + X_i'\gamma + \lambda_t + \varepsilon_{it}, \tag{1}$$

All density variables in the equations denote inverse-hyperbolic-sine transformed values, calculated from quantities per square kilometer; geological shares remain untransformed. The year effects, λ_t, allow average density to differ across years. The first specification includes bedrock share and year effects. The preferred specification adds X_i, which contains controls for location and the area of each geographical unit.

Because bedrock is fixed over time, β is estimated from differences between areas. Pooling the historical observations estimates a common geological association across years. I also run the regression separately for each year to examine how that association changes over time. The population regressions use the same 381 areas at every date; the mantal regressions use the available observations at each date.

The location controls are a full cubic polynomial in the coordinates of a representative point inside each historical area.[2] These controls account for broad geographical differences in density across Skåne, such as those between its northern and southern parts. The geological coefficient is then estimated from variation remaining after these regional patterns have been taken into account. I also control for the area of each parish or city.

The coefficient β corresponds to a change in carbonate-rich share from zero to one. A ten-percentage-point difference in that share corresponds to a fitted difference of 0.1β in transformed density. Since the regression includes a single geological share, the comparison is with the mix of other bedrock types found in Skåne.

Bedrock was formed long before the historical economy, ruling out reverse causality from agriculture and settlement to geological composition. The regressions account for broad geographical differences through flexible location controls and area size. Additional comparisons introduce county controls and exclude areas around major cities, examining whether the historical relationship persists across administrative divisions and beyond urban centers.

4.2 Settlement and Modern Economic Activity

For the settlement records, I estimate

$$s_i = \alpha + \beta G_i + u_i. \tag{2}$$

Here s_i denotes settlement-record density. I estimate the equation separately for church and town records and for the broader settlement measure. These regressions describe the cross-area association with bedrock without location or area controls.

Towns could serve agricultural areas extending beyond their own boundaries. I therefore also compare total town population in 1810 with rural agricultural capacity within 10, 20, and 30 kilometers of each historical core. Scatter plots and unweighted linear fits show how this relationship changes with the extent of the surrounding countryside included. These comparisons use population and mantal in levels.

I next examine whether places that were densely settled in the past also concentrate people and economic activity today. For each modern density outcome y_i, I estimate

$$y_i = \alpha + \beta_\tau G_i + \theta_\tau q_{i,\tau} + X_i'\gamma + v_i, \tag{3}$$

[2]Coordinates are measured in kilometers and centered before constructing the polynomial. It includes x, y, x^2, xy, y^2, x^3, x^2y, xy^2, and y^3.

The main outcomes are population, workplaces, employment, and value added per square kilometer. For modern population, the historical year τ is 1570, 1750, or 1810. The workplace outcomes use 1810. The geographical sensitivity checks also include employed-resident density. For each outcome, I estimate a specification without historical population to show the geological association before accounting for the inherited population pattern.

For modern population, I compare the explanatory power of population density in 1570, 1750, and 1810 using the same 381 areas and the same geographical controls. I also include all three historical years together and compare specifications that omit one or both earlier years. This shows whether the earlier population records add information beyond the 1810 map.

A positive θ_τ indicates that historically more densely populated areas also have greater modern density, conditional on geology and the geographical controls. This persistence may reflect several advantages accumulated in established settlements, including roads, markets, and institutions.

Adding historical population also changes the interpretation of the geological coefficient. Let b_0 be its coefficient without historical population and b_1 its coefficient when $q_{i,\tau}$ is included. Let π_τ be the coefficient on bedrock when that historical population measure is regressed on bedrock and the same controls. With the same observations and controls in all three regressions, ordinary least squares gives

$$b_0 - b_1 = \theta_\tau \pi_\tau. \tag{4}$$

The difference $b_0 - b_1$ measures how much of the geological association is shared with historical population.

For example, agricultural conditions may have supported both a town and the roads serving it. Those roads could have attracted residents in 1810 and firms in later periods. Historical population would then reflect part of the inherited transport advantage, while some could remain associated with geology even after population is included. Measurement error in historical population can also leave a residual geological coefficient. The reduction in that coefficient is therefore informative about shared historical geography, but assigning it to a particular causal channel requires additional assumptions.[3]

[3] Kwon and Roth (2026) develop tests of full mediation under explicit assumptions about treatment assignment and the relationship between treatment, the mediator, and the outcome. Applying such tests here would require a model of geological exposure and historical population beyond the regressions reported.

4.3 Magnitudes and Inference

For modern population density, I report standard-deviation and interquartile contrasts to express the estimates on a common scale. These are useful because coefficients on the transformed densities are not generally elasticities, especially when densities are low. Changes in R^2 show how much additional variation in the modern outcome is accounted for by historical population.

I report standardized coefficients and interquartile contrasts for bedrock share and historical population density, both expressed in standard deviations of the dependent variable. All calculations use the variables as entered in the regressions and the common sample of 381 areas.

Historical panel regressions use standard errors clustered by parish or city, allowing errors for the same area to be correlated across years. Cross-sectional regressions use heteroskedasticity-robust standard errors. To examine sensitivity to correlation between nearby areas, the appendix reports Conley-style standard errors over several distance cutoffs for the baseline historical and modern residential regressions. These spatial checks do not cover the settlement regressions, models including all three historical population measures, specifications with county controls, or workplace regressions. The appendix also compares the pooled historical coefficient with a distribution obtained by randomly reassigning bedrock shares across parishes, showing how unusual the observed association is relative to those reshuffled allocations.

5 Bedrock and the Historical Economy

The agricultural interpretation predicts that areas with more carbonate-rich bedrock had greater agricultural capacity and denser populations. I examine this prediction using the population and mantal records, before turning to the evidence on churches, towns, and other settlements.

5.1 Population and Agricultural Capacity

Table 2 reports the estimates of equation (1), pooling observations from the five historical years. Columns (1) and (3) include year effects, allowing average density to differ across years. Columns (2) and (4) add the cubic coordinate polynomial and controls for the size of each historical area. These regressions extend the comparison in Figure 2 to the full historical record and to agricultural capacity.

Columns (1) and (2) show a strong positive association between carbonate-rich bedrock and historical population density. The coefficient is 0.483 in column (1) and remains positive and statistically significant at the 1 percent level when the full cubic coordinate polynomial and area controls are included in column (2): 0.179 (standard error 0.060). Areas with a larger share of carbonate-rich bedrock were thus more densely populated even after accounting for broad geographical differences within Skåne.

Columns (3) and (4) show a strong positive association with agricultural capacity, measured by mantal per square kilometer. Mantal records the assessed taxable capacity of agricultural holdings. The coefficient on carbonate-rich share is 0.394 in column (3) and remains positive and statistically significant at the 1 percent level with the same location and area controls in column (4): 0.141 (standard error 0.043). The population relationship therefore has a clear counterpart in the agricultural economy: areas with more carbonate-rich bedrock had both denser populations and greater agricultural capacity per square kilometer.

Table 2: Bedrock, population, and agricultural capacity, 1570–1810

Dependent variable:	Population density		Agricultural capacity (mantal density)	
	(1)	(2)	(3)	(4)
Carbonate-rich bedrock share	0.483^{***}	0.179^{***}	0.394^{***}	0.141^{***}
	(0.045)	(0.060)	(0.034)	(0.043)
Observations	1,905	1,905	1,894	1,894
Historical areas	381	381	381	381
R^2	0.42	0.65	0.36	0.58
Location controls	No	Yes	No	Yes
Area control	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Notes: Pooled regressions for 1570, 1630, 1690, 1750, and 1810. Eleven mantal observations are missing. Density variables are measured per square kilometer and transformed as described in Section 4; bedrock share is on a zero-to-one scale. Location controls are a full cubic coordinate polynomial. Standard errors in parentheses are clustered by historical area.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

This correspondence is central to the agricultural interpretation. Land capable of supporting more farming households could also generate demand for trade, religious services, and administration in nearby villages and towns. The positive associations with both mantal and population are consistent with agriculture supporting a larger local economy. Since mantal also reflects assessment practices and the organization of holdings, the county comparisons in Section 7 examine whether broad administrative differences account for these results.

The year-by-year estimates in Table 3 show a stable positive relationship over the 240 years from 1570 to 1810. With the same location and area controls throughout, the population coefficient varies only between 0.161 and 0.199 and is statistically significant at the 5 percent level in all five years. The corresponding mantal coefficients are positive throughout and range from 0.120 to 0.167. The population regressions use exactly the same 381 areas in each year; the mantal sample ranges from 375 to 381 areas.

Table 3: Bedrock, population, and agricultural capacity by year

Year	1570	1630	1690	1750	1810
Panel A. Population density					
Carbonate-rich bedrock share	0.196^{***}	0.163^{**}	0.172^{***}	0.161^{***}	0.199^{***}
	(0.069)	(0.067)	(0.065)	(0.061)	(0.070)
Observations	381	381	381	381	381
R^2	0.61	0.58	0.46	0.48	0.54
Panel B. Agricultural capacity (mantal density)					
Carbonate-rich bedrock share	0.120^{*}	0.134^{***}	0.145^{**}	0.139^{***}	0.167^{***}
	(0.068)	(0.051)	(0.060)	(0.048)	(0.042)
Observations	381	376	381	381	375
R^2	0.55	0.43	0.43	0.55	0.61
Location controls	Yes	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes	Yes

Notes: Each column reports separate regressions for the year shown. Population uses the same 381 historical areas throughout; mantal uses the available observations. Variable definitions and controls follow Table 2. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The population coefficient is almost identical at the beginning and end of the historical period: 0.196 in 1570 and 0.199 in 1810. The relationship persists across Skåne's transfer from Danish to Swedish rule. Areas with more carbonate-rich bedrock were thus already more densely populated centuries before Sweden's broad industrial transformation and remained so throughout the historical record.

5.2 Settlement and Durable Places

Church and town records identify centers of religious, commercial, and administrative activity in the agrarian economy. Heritage records and data on place names provide a broader picture of settlement. I examine whether these places were more concentrated in areas with carbonate-rich bedrock.

Table 4 compares the two measures defined in Section 3.2. Column (1) uses church and bell-tower records with histories extending to 1810 or earlier and indicators for ten historical towns. Column (2) adds mapped heritage features and selected names, which cover a wider range of local activity and do not share the same date restriction. Both counts are expressed per square kilometer, using the same 381 historical areas. The table reports cross-area associations without location or area controls.

Areas with more carbonate-rich bedrock contain more recorded churches, towns, and other settlement features per square kilometer. The carbonate-rich coefficient is 0.032 (standard error 0.007) for churches and towns and 0.323 (0.053) for the broader measure. Both estimates are statistically significant at the 1 percent level.

Table 4: Bedrock and settlement records

Dependent variable	(1) Church and town record density	(2) Broader settlement record density
Carbonate-rich bedrock share	0.032^{***}	0.323^{***}
	(0.007)	(0.053)
Observations	381	381
R^2	0.07	0.09
Location controls	No	No
Area control	No	No

Notes: Both columns use the same 381 historical areas. Column (1) counts 341 church records and two bell-tower records with recorded histories extending to 1810 or earlier, plus ten historical-town indicators. Column (2) adds 20,123 mapped heritage features and 5,129 selected names, without a common date restriction. Counts are divided by historical area in square kilometers; entries need not identify distinct settlements. Section 3.2 and Appendix A.1 give the definitions and selection rules. Density variables are transformed as described in Section 4; bedrock share is on a zero-to-one scale. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The records place religious and urban centers, together with a broader range of recorded settlement features, within the same geological geography as agricultural capacity and population. A center could also serve farms beyond its own boundaries. The next comparison examines this wider agricultural setting directly.

5.3 Agricultural Districts and Population Centers

Agricultural capacity and population could be concentrated in different parts of the same district. A town might contain many residents and little farmland, while nearby parishes supplied produce and customers for its markets, crafts, and services. This suggests comparing

the size of a town with the agricultural capacity of its hinterland, as well as examining conditions within individual parishes.

Figure 3 relates population in the ten historical town areas in 1810 to rural mantal within 10, 20, and 30 kilometers of their historical cores. Larger towns tend to have greater agricultural capacity in the wider surrounding countryside. The correlation is 0.10 within 10 kilometers, rising to 0.50 at 20 kilometers and 0.64 at 30 kilometers. The 30-kilometer relationship remains positive when any one town is omitted, with correlations between 0.57 and 0.69.

The positive relationship between town population and agricultural capacity in the surrounding countryside is consistent with the hinterland mechanism developed by Nagy (2023). Total capacity reflects both the extent of rural land and its assessed capacity per unit of area.

Population records capture the concentration of people supported by this wider economy, including those engaged in trade and services. They therefore provide a natural link to the modern comparisons. Buildings, markets, roads, and local services accumulated in established settlements could remain useful as employment moved into industry and services. By the earliest observations, agricultural capacity and population already shared a clear geographical pattern. The next section examines how much of the historical population geography remains visible in the distribution of modern residents and workplaces.

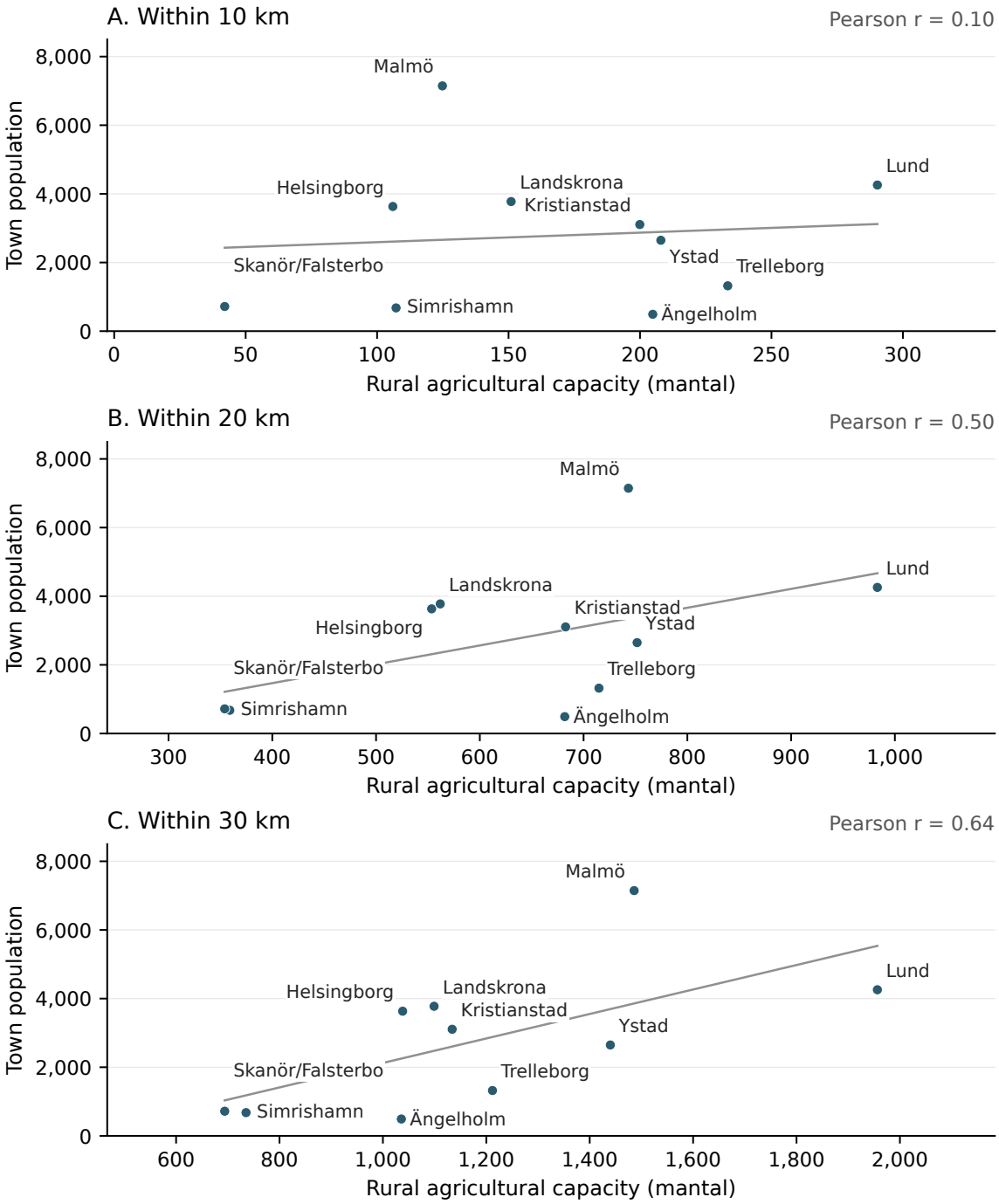


Figure 3: Town population and agricultural hinterlands, 1810

Notes: Each point represents one historical town area. Population and mantal are measured in 1810. Rural mantal is summed within the indicated distance of historically documented central church sites, using parish-area overlap weights and excluding all city areas. Skanör and Falsterbo form one observation; their catchments are combined without double counting. Capacity covers rural parishes in the Skåne study region. Lines are unweighted linear fits. Sources: Sweden 1570–1810 through SND, Lantmäteriet, and Bebyggelseregistret.

6 Persistence into the Modern Economy

The historical results show that carbonate-rich areas supported greater agricultural capacity and denser populations. I now examine whether this population geography persists in the modern distribution of residents and economic activity. The workplace outcomes are important here: historically populous areas may remain attractive places to live even if employment and production have moved elsewhere.

6.1 The Historical Population Map

The 1810 population map provides the starting point. It records the distribution of people before Sweden's broad industrial transformation and the expansion of railways. I first relate this map to modern residential density, then use the earlier population records to examine how far back the relationship can be traced. All comparisons use the same 381 historical areas and the same location and area controls.

Column (1) of Table 5 shows that carbonate-rich bedrock is positively associated with population density in 2025, conditional on location and area. The coefficient is 0.461 and significant at the five-percent level. Column (4) adds population density in 1810, which enters with a coefficient of 1.083, significant at the one-percent level.

Adding population density in 1810 reduces the coefficient on carbonate-rich bedrock from 0.461 to 0.245, a decline of almost half. Much of the modern geological association is thus shared with a population geography established before industrialization. This pattern is consistent with agricultural advantages supporting settlements whose housing, infrastructure, and market connections continued to attract residents as the economy changed.

The reduction connects the historical and modern results. Carbonate-rich bedrock is associated with higher population density in 1810, and historically populous areas remain more densely populated today. By equation (4), the decline in the bedrock coefficient when population density in 1810 is included equals the product of these two coefficients, $0.199 \times 1.083 \approx 0.216$.

The earlier population records trace this persistence back to the sixteenth century. Replacing the 1810 measure with population density in 1570 or 1750 yields coefficients of 0.791 and 0.984 in columns (2) and (3), both significant at the one-percent level. Areas that were densely settled in 1570 remain more densely populated today, even after accounting for bedrock and the geographical controls.

The 1570 map also provides much of the gain in fit obtained with 1810. Relative to the specification with bedrock and geographical controls in column (1), adding population

Table 5: Historical population and modern residential density

Dependent variable:	Population density, 2025				
	(1)	(2)	(3)	(4)	(5)
Historical population year	None	1570	1750	1810	All three
Carbonate-rich bedrock share	0.461**	0.306	0.302	0.245	0.238
	(0.222)	(0.210)	(0.213)	(0.208)	(0.210)
Population density, 1570		0.791***			0.212
		(0.200)			(0.238)
Population density, 1750			0.984***		0.365
			(0.215)		(0.282)
Population density, 1810				1.083***	0.616**
				(0.225)	(0.296)
Observations	381	381	381	381	381
R^2	0.28	0.33	0.34	0.34	0.35
Adjusted R^2	0.26	0.30	0.31	0.32	0.32
Location controls	Yes	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes	Yes

Notes: All regressions use the same 381 historical areas. Densities are measured per square kilometer and transformed as described in Section 4; bedrock share is on a zero-to-one scale. Location controls are a full cubic coordinate polynomial. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

density in 1570 raises R^2 from 0.277 to 0.325, a gain of 4.8 percentage points. Using 1810 raises R^2 to 0.342, a gain of 6.5 percentage points.

Combining all three historical years in column (5) yields little further improvement over the 1810 specification. The coefficients on population in 1570 and 1750 become smaller and less precise, while the 1810 coefficient remains positive and significant. The three historical maps therefore appear to capture much of the same persistent settlement geography, with 1810 providing the best single-year fit for modern population density.

6.2 Economic Magnitudes

The estimates imply substantial differences in modern population density. Table 6 reports standardized coefficients and interquartile contrasts. Moving from the twenty-fifth to the seventy-fifth percentile of population density in either 1570 or 1810 corresponds to about 0.44 standard deviations of modern population density, conditional on bedrock share, location, and area. The size of this contrast is strikingly similar for historical observations separated by more than two centuries.

The standardized coefficient on historical population density is 0.348 in 1570 and 0.373 in 1810. The standardized bedrock coefficient, before conditioning on historical population, is 0.154. Historical population remains a strong predictor of modern population density, conditional on bedrock, location, and area.

Table 6: Standardized associations with modern residential density

Dependent variable:	Population density, 2025	
Increase in predictor:	1 SD	P25–P75
Carbonate-rich bedrock share	0.154	0.338
Population density, 1570	0.348	0.444
Population density, 1750	0.334	0.358
Population density, 1810	0.373	0.445
Observations	381	381

Notes: Entries are fitted differences in standard deviations of the dependent variable, for a one-standard-deviation or interquartile increase in the predictor. Calculations use the variables as entered in Table 5: column (1) for bedrock and columns (2)–(4) for historical population. All specifications control for location and area; historical-population estimates also condition on bedrock share.

Comparing the Scanian results with evidence from other settings helps assess the extent of persistence. An important benchmark is the study of Japan by Davis and Weinstein (2002), who document the durability of regional population patterns over centuries of economic change. For population density in 1600 and 1998, they find a raw correlation of 0.76 and a rank correlation of 0.83. Calculating corresponding correlations for Skåne provides a common descriptive basis for comparison.

I therefore correlate each area's historical population density with its density in 2025, using the same 381 historical areas. The Pearson correlation is 0.48 between 1570 and 2025, 0.46 between 1750 and 2025, and 0.51 between 1810 and 2025. The corresponding Spearman rank correlations are 0.30, 0.23, and 0.34. The calculations use untransformed population per square kilometer, include no controls, and give each area equal weight. The similar Pearson correlations for 1570 and 1810 show that the relationship with modern population density extends back well before industrialization.

The correlations are lower than those for Japan, particularly for ranks. However, the geographical scale differs considerably: the Japanese comparison covers 39 large regions across the country, whereas the Scanian analysis uses 381 smaller areas within a single region.

6.3 Workplaces, Employment, and Value Added

I next examine whether historically populous areas also concentrate establishments, employment, and production today. An inherited housing stock and local amenities may sustain residential concentrations even when people commute elsewhere for work. Persistence in these outcomes would show that the historical geography extends to where economic activity takes place, as well as where people live.

Table 7 reports results for average workplace, employment, and value-added densities over 1997–2021. Panel A asks whether the geological characteristic associated with historical agricultural capacity also predicts the location of modern economic activity. Conditional on location and area, carbonate-rich share enters positively in all three regressions, with coefficients of 0.317, 0.813, and 1.133. Panel B adds population density in 1810 to examine how much of this relationship is captured by the settlement pattern established before industrialization. The coefficients on historical population are 0.956 for workplaces, 1.271 for employment, and 1.338 for value added, all significant at the one-percent level.

Table 7: Historical population and modern economic activity, 1997–2021

Dependent variable:	(1) Workplace density	(2) Employment density	(3) Value-added density
Panel A. Without historical population			
Carbonate-rich bedrock share	0.317^{**}	0.813^{***}	1.133^{**}
	(0.160)	(0.232)	(0.469)
Observations	381	381	381
R^2	0.27	0.23	0.12
Panel B. With historical population			
Carbonate-rich bedrock share	0.126	0.560^{***}	0.866^{*}
	(0.141)	(0.209)	(0.469)
Population density, 1810	0.956^{***}	1.271^{***}	1.338^{***}
	(0.171)	(0.269)	(0.275)
Observations	381	381	381
R^2	0.38	0.31	0.15
Location controls	Yes	Yes	Yes
Area control	Yes	Yes	Yes

Notes: Statistics Sweden's establishment records cover the private sector. Workplace density counts establishments per square kilometer; employment and value added are totals per square kilometer. Annual densities are averaged over the years observed within 1997–2021, then transformed as described in Section 4. All regressions use the same 381 historical areas. Location controls are a full cubic coordinate polynomial. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Including historical population reduces the geological coefficients to 0.126, 0.560, and 0.866. For workplace density, the remaining estimate is imprecise; the employment coefficient remains significant at the one-percent level, and the value-added coefficient at the ten-percent level. The decline shows that the inherited population pattern helps account for the concentration of modern activity in carbonate-rich areas. Together with the strong historical-population coefficients, this is consistent with settlements supported by the agricultural economy retaining their importance as locations of employment and production.

The outcomes capture different aspects of local activity. Employment density depends on both the number and size of establishments; value-added density also reflects sector composition and production per worker. Access to workers, suppliers, and transport networks may help established centers attract larger employers or particular industries. Population density in 1810 may capture these inherited advantages only imperfectly, which could explain why employment and value added remain related to bedrock after historical population is included.

The link between historical population and modern employment is also documented by Combes et al. (2010, Table 1.2). Across 306 French employment areas, they report correlations of 0.75 and 0.88 between log urban population density in 1831 and 1881, respectively, and log employment density during 1976–1996. For Skåne, the unconditional Pearson correlation between log population density in 1810 and log mean annual private-sector employment density over 1997–2021 is 0.36, with equal weights for the same 381 historical areas. This is lower than the French correlations. Their national comparison uses larger labor-market areas and historical population measured only in towns, however, whereas the Scanian calculation includes rural populations and compares smaller parish and city areas within a single region.

Appendix Table A2 reports shorter time windows and additional monetary outcomes. The historical population coefficients remain positive across the windows, and the conditional workplace-count relationship with geology remains imprecise. The association with the historical population map is therefore present in both the full-period averages and the more recent observations.

The historical population geography persists in both residence and the location of production. Areas that supported substantial agrarian populations continue to attract establishments and employment, linking the agricultural economy to the places where modern economic activity is carried out.

7 Geography and Spatial Inference

The preceding results suggest that agricultural conditions helped shape a settlement geography that remains visible today. Institutional differences between counties could have shaped tax assessment practices, so part of the historical relationship between bedrock and assessed agricultural capacity might reflect differences in administration rather than agricultural conditions. I therefore examine whether this relationship remains within counties. I then ask whether the historical relationships between bedrock, agricultural capacity, and population extend beyond the major cities, and whether the corresponding relationship with modern population is concentrated around these centers. To do so, I compare the full-sample estimates with those obtained after excluding areas near major cities. I also account for proximity to Copenhagen, since access to a large nearby market could contribute to the modern concentration of activity. Finally, neighboring areas share markets, infrastructure, and physical conditions that may correlate their regression errors. I therefore examine whether allowing for spatial dependence changes the precision of the estimates.

7.1 County Differences

Skåne was divided into two counties, Malmöhus and Kristianstad. Institutional differences between them could have affected the assessment of agricultural holdings for taxation. If these differences coincided with geology, part of the estimated relationship between bedrock and mantal could reflect assessment practices rather than agricultural conditions. I therefore add county fixed effects alongside the baseline location and area controls, comparing areas within the same county.

The two counties' boundaries were broadly stable between 1719 and 1996 (Linde and Andersson Palm, 2014, pp. 9, 83, 89). I therefore restrict the historical comparison to 1750 and 1810. County membership follows the harmonized classification in these sources rather than a separate reconstruction of boundaries for each year.

Table 8 shows that county fixed effects leave the estimates almost unchanged. The population coefficient moves from 0.161 to 0.157 in 1750 and from 0.199 to 0.198 in 1810; the corresponding mantal coefficients move from 0.139 to 0.133 and from 0.167 to 0.164. All remain positive and statistically significant. The historical relationships therefore reflect differences within the two counties, rather than simply a contrast between their average population or assessed agricultural capacity.

Columns (3)–(4) exclude eleven parishes whose recorded county affiliation in 1742 is split or differs from the harmonized classification. The estimates are again very similar, indicating that these discrepancies do not drive the results.

Table 8: Bedrock, historical density, and county fixed effects

	Full sample		Excluding 11 parishes	
	(1)	(2)	(3)	(4)
Panel A. Population density, 1750				
Carbonate-rich bedrock share	0.161***	0.157**	0.169***	0.163***
	(0.061)	(0.062)	(0.061)	(0.063)
Observations	381	381	370	370
R^2	0.48	0.48	0.49	0.49
Panel B. Population density, 1810				
Carbonate-rich bedrock share	0.199***	0.198***	0.202***	0.198***
	(0.070)	(0.070)	(0.071)	(0.072)
Observations	381	381	370	370
R^2	0.54	0.54	0.54	0.54
Panel C. Agricultural capacity (mantal density), 1750				
Carbonate-rich bedrock share	0.139***	0.133***	0.134***	0.127**
	(0.048)	(0.049)	(0.049)	(0.050)
Observations	381	381	370	370
R^2	0.55	0.56	0.55	0.56
Panel D. Agricultural capacity (mantal density), 1810				
Carbonate-rich bedrock share	0.167***	0.164***	0.163***	0.159***
	(0.042)	(0.042)	(0.043)	(0.043)
Observations	375	375	364	364
R^2	0.61	0.61	0.61	0.61
Location controls	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes
County fixed effects	No	Yes	No	Yes

Notes: Panel headings identify the dependent variable and year. Location controls are a full cubic coordinate polynomial. County fixed effects distinguish Malmöhus and Kristianstad using the historical sources' harmonized classification. Columns (3)–(4) exclude eleven parishes whose recorded 1742 county affiliation is split or differs from that classification. Six mantal observations are missing in 1810. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

County fixed effects also leave the historical-to-modern relationship essentially unchanged (Appendix Table A3). The coefficient on population density in 1810 changes from 1.083 to 1.084 for modern population and from 1.062 to 1.063 for employed residents. Persistence is thus evident within the historical counties as well as across Skåne as a whole.

7.2 Cities and Access to Copenhagen

Did the historical agricultural relationship extend beyond the major cities, and does the same hold today? Table 9 compares the baseline estimates with specifications that control for distance to Copenhagen, exclude areas within 10 kilometers of the six major city centers, eexclude the historical areas corresponding to towns that existed before 1810, or exclude areas within 30 kilometers of Copenhagen. All use the baseline location and area controls. Appendix Figure A2 maps the cities, towns, and areas within 10 kilometers of the major city centers.

The historical relationship clearly extends beyond the areas surrounding major cities. Excluding the 10-kilometer neighborhoods in column (3) leaves a population coefficient of 0.159, compared with 0.179 in the full panel, and a mantal coefficient of 0.125, compared with 0.141. Both remain significant at the five-percent level. The separate population regression for 1810 gives the same pattern. Areas with more carbonate-rich bedrock thus supported greater agricultural capacity and denser populations even outside the major-city neighborhoods.

In contrast, the modern relationship between bedrock and residential density is concentrated around major cities. Excluding areas within 10 kilometers of the city centers reduces the coefficients from 0.461 to 0.029 for population and from 0.470 to 0.028 for employed residents. Excluding only the ten historical areas corresponding to towns that existed before 1810 leaves larger coefficients of 0.314 and 0.324, respectively. Although both sets of estimates are imprecise, the sharper decline under the broader exclusion—which removes 80 of the 381 areas—suggests that the modern relationship involves the cities and their surroundings, rather than the historical town areas alone.

The results are consistent with agricultural districts sustaining towns whose markets, buildings, and transport connections remained valuable as employment shifted toward industry and services. This contrast suggests that the legacy of agricultural geography is now concentrated around urban centers, even though the historical relationship extended more broadly across the countryside.

Proximity to Copenhagen provides another possible source of modern concentration. Controlling for distance to Copenhagen reduces the modern bedrock coefficients to 0.367

Table 9: Bedrock and density: city exclusions and Copenhagen

	Baseline (1)	Copenhagen distance (2)	Outside city neighborhoods (3)	Excluding pre-1810 towns (4)	Outside Copenhagen 30 km (5)
Panel A. Population density, 1570–1810					
Carbonate-rich share	0.179***	0.151**	0.159**	0.115**	0.173***
	(0.060)	(0.065)	(0.065)	(0.048)	(0.062)
Observations	1,905	1,905	1,505	1,855	1,860
R^2	0.65	0.65	0.70	0.71	0.65
Panel B. Agricultural capacity (mantal density), 1570–1810					
Carbonate-rich share	0.141***	0.097**	0.125**	0.155***	0.160***
	(0.043)	(0.045)	(0.057)	(0.039)	(0.042)
Observations	1,894	1,894	1,498	1,855	1,851
R^2	0.58	0.58	0.57	0.62	0.59
Panel C. Population density, 1810					
Carbonate-rich share	0.199***	0.144**	0.186**	0.124**	0.183***
	(0.070)	(0.071)	(0.082)	(0.055)	(0.066)
Observations	381	381	301	371	372
R^2	0.54	0.54	0.62	0.62	0.55
Panel D. Population density, 2025					
Carbonate-rich share	0.461**	0.367	0.029	0.314	0.493**
	(0.222)	(0.250)	(0.241)	(0.209)	(0.228)
Observations	381	381	301	371	372
R^2	0.28	0.28	0.27	0.29	0.27
Panel E. Employed-resident density, 2024					
Carbonate-rich share	0.470**	0.406*	0.028	0.324	0.507**
	(0.218)	(0.245)	(0.233)	(0.204)	(0.223)
Observations	381	381	301	371	372
R^2	0.30	0.30	0.30	0.32	0.29
Location controls	Yes	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes	Yes

Notes: Entries are coefficients on carbonate-rich bedrock share; panel headings identify the dependent variables. Panels A–B pool the five historical years and include year fixed effects. Location controls are a full cubic coordinate polynomial. Column (2) adds $\log(1+d)$, where d is distance to Copenhagen in kilometers. Columns (3)–(5) exclude, respectively, areas within 10 km of Malmö, Lund, Helsingborg, Kristianstad, Ystad, or Trelleborg; pre-1810 town parishes; and areas within 30 km of Copenhagen. Distances are straight-line distances from each area's representative interior point. Employed residents are aged 20–64. Standard errors in parentheses are clustered by historical area in Panels A–B and heteroskedasticity-robust in Panels C–E. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and 0.406. Excluding the nine areas within 30 kilometers instead leaves coefficients of 0.493 and 0.507, both significant at the five-percent level. The modern relationship is therefore not confined to the areas nearest Copenhagen.

7.3 Spatial Dependence

Nearby parishes share physical geography, markets, and historical institutions, so their regression errors may be correlated. Parish clustering allows dependence across years within an area but not across neighboring areas. Appendix Tables A4 and A5 therefore report spatial HAC standard errors (Conley, 1999) using distance cutoffs of 25, 50, and 100 kilometers. These cover the baseline historical bedrock regressions, pooled and for 1810, and the regressions relating modern residential density to population in 1810.

For the pooled historical population coefficient of 0.179, the standard error rises from 0.060 with parish clustering to 0.092 at the 50-kilometer cutoff. The mantal coefficient of 0.141 has a standard error of 0.054 at that cutoff. Spatial adjustment therefore reduces precision more for historical population than for agricultural capacity. The relationship between population in 1810 and modern population remains significant at the one-percent level at all three cutoffs, as does the relationship with employed-resident density.

Appendix Table A4 also compares the actual results with those obtained when bedrock shares are randomly reassigned across parishes. I shuffle the observed carbonate-rich shares 999 times and re-estimate the pooled population and mantal regressions after each shuffle. Each parish retains its observed outcomes and controls, and its reassigned bedrock share is held fixed across years. For both population and mantal, fewer than 1 percent of these reassignments yield a coefficient as large in absolute value as the actual estimate. The result means that coefficients of the observed size rarely arise when the same bedrock shares are randomly assigned to different places.

8 Conclusion

The economic geography of modern Skåne retains a clear imprint of its agrarian past. Areas with more carbonate-rich bedrock had greater assessed agricultural capacity and denser populations throughout the historical record from 1570 to 1810. Observing these two outcomes together connects the inherited population map to the agricultural economy that supported it. The stability of the relationship across Danish and Swedish records shows that this connection endured through changes in state authority and fiscal administration.

Records of churches, towns, and other settlement features are also more concentrated in carbonate-rich areas. In 1810, towns with larger populations tended to have greater total agricultural capacity in the wider surrounding countryside. These findings are consistent with agricultural districts sustaining rural communities as well as nearby centers of trade and services.

The relationship with modern population is already present in the records for 1570, centuries before Sweden's broad industrial transformation. Although the 1810 population map provides the best single-year fit, an interquartile increase in either historical measure corresponds to about 0.44 standard deviations in modern population density, conditional on bedrock, location, and area. The similar magnitudes show that the earlier population map contains economically substantial information about where people live today.

Persistence also extends to the location of production. Population density in 1810 predicts establishment, employment, and value-added densities over 1997–2021. Historically populous areas therefore remain economically active places, rather than merely residential concentrations sustained by an inherited housing stock or local amenities.

The geographical reach of the bedrock relationship has nevertheless changed. Historically, the relationships with population and agricultural capacity extend beyond the areas around major cities. Today, the relationship with residential density is concentrated around those centers. This contrast is consistent with agricultural districts supporting towns whose markets, buildings, transport connections, and labor pools remained valuable as employment shifted toward industry and services. The wider implication is that an agricultural advantage can persist through the places it helped establish, even as the economic role of the land itself changes.

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A Data Sources and Definitions

Table A1: Data sources and measurement

Measure	Source	Measurement
Carbonate-rich bedrock	Geological Survey of Sweden	Share of each historical parish or city underlain by carbonate-rich bedrock.
Historical population and agricultural capacity (mantal), 1570–1810	Swedish National Data Service parish panel	Population and mantal totals aggregated within each historical area and divided by its area.
Church and town records	Bebyggelseregistret and historical towns identified in Lantmäteriet's city boundaries	Religious-building records with histories extending to 1810 or earlier, plus one indicator per historical town, divided by historical area.
Broader settlement records	The preceding sources, Fornsök, and Lantmäteriet Ortnamn	The church and town count plus selected mapped heritage features and names, divided by historical area. Selection rules are given in Appendix A.1.
Modern population and employed residents	Statistics Sweden small-area statistics (DeSO)	Counts allocated according to the share of each DeSO area covered by a historical area, then expressed per square kilometer.
Workplaces, employment, and value added	Statistics Sweden, restricted-access private-sector workplace records	Workplace coordinates identify their historical area. Annual workplace counts, employment, and value added are summed within each area, divided by its area, and averaged over the relevant period.

Notes: Historical parish and city boundaries are from Lantmäteriet. The main matched sample contains 381 areas. Mantal measures assessed agricultural capacity.

A.1 Settlement records

Table 4 uses two measures: religious-building and town records (353 entries), and a broader measure that adds heritage features and selected names (25,605 entries). Both are expressed per square kilometer of the full parish or city area and cover the same 381 historical areas.

Religious buildings and towns. I select church, chapel, funerary-chapel, and bell-tower records from Bebyggelseregistret if their earliest recorded historical event or building phase

begins by 1810. Keeping one record per building identifier gives 341 churches and two bell towers. These dates describe building histories, not necessarily the founding of settlements.

The ten town areas, defined by Lantmäteriet's city boundaries, are Helsingborg, Kristianstad, Landskrona, Lund, Malmö, Simrishamn, Skanör med Falsterbo, Trelleborg, Ystad, and Ängelholm. Each contributes one town indicator; Skanör and Falsterbo form a single area.

Heritage features. The broader measure adds 20,123 mapped features from Fornsök: occupation sites, village and farm plots, croft remains, house foundations, hearths and cooking pits, religious remains, mills, old routes, and cup-mark rock carvings.

I search the site-type and feature descriptions for *boplats*, *boplatsområde*, *boplatslämning*, *härd*, *kokgrop*, *husgrund, förhistorisk/medeltida*, *grav- och boplatsområde*, *bytomt*, *gårdstomt*, *lägenhetsbebyggelse*, *husgrund, historisk tid*, *kyrka/kapell*, *kloster*, *kvarn*, and *färdväg*. The search ignores capitalization and includes matches within longer descriptions. There is no restriction on age, recording method, or protection status.

Names. Lantmäteriet's Ortnamn database contributes 5,129 entries: church names and selected building or settlement names. Names classified as buildings or settlements qualify if they end in *-by*, *-torp*, *-arp*, *-rup*, *-löv*, *-lev*, *-stad*, *-tofta*, *-inge*, *-åkra*, *-hult*, *-ryd*, *-rum*, *-lunda*, or *-lösa*; contain *gård*, *gården*, or *gard*; or end in *säteri*, *herrgård*, *slott*, *kungsgård*, *sätesgård*, *herresäte*, *gods*, or *borg*. Selection uses names in the modern register without a historical date restriction.

Location and counting. Entries are assigned to the historical area containing their coordinates; lines and polygons are represented by interior points. A feature or name matching several search terms is counted once. The 20,123 heritage features correspond to 19,167 register identifiers, since one record can contain several mapped features. Records from different sources, including church records and town indicators, remain separate even when they describe the same place.

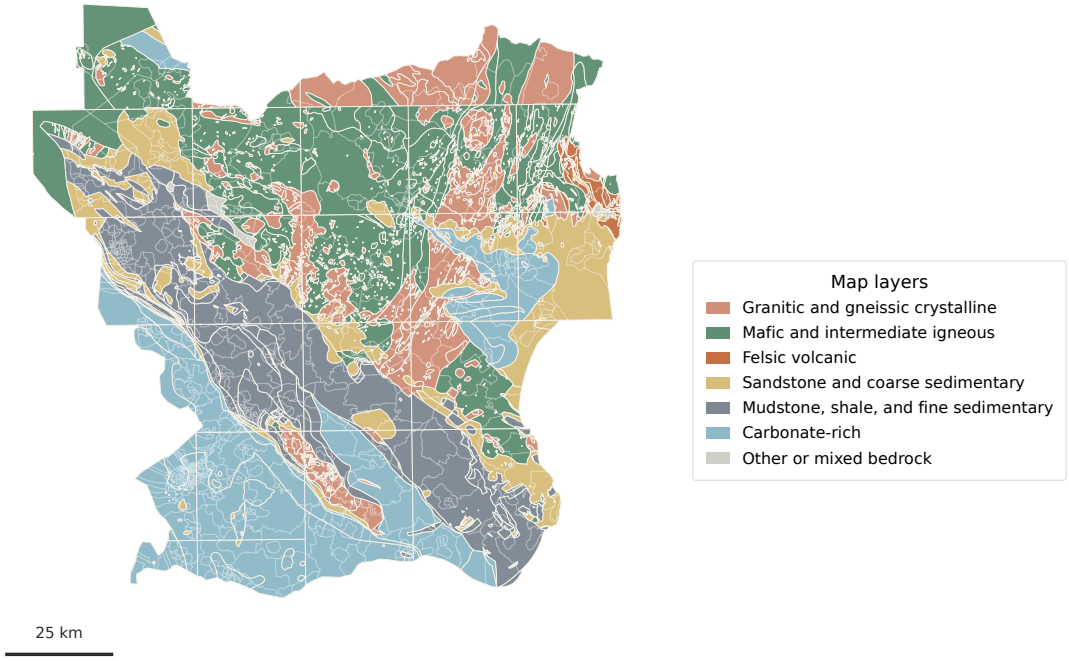


Figure A1: Bedrock geology in Skåne

Notes: Colors show the bedrock groups used in the analysis. Bedrock data are from SGU; the map outline and local boundaries are from Statistics Sweden.

B Modern Economic Activity

Table A2: Modern economic activity across time periods

Dependent variable	Workplaces		Employment		Value added		Production value		Labor costs		Raw-material costs	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A. 1997–2021												
Carbonate-rich bedrock	0.32**	0.13	0.81***	0.56***	1.13**	0.87*	0.69***	0.40*	0.86***	0.57**	0.80***	0.50*
	(0.16)	(0.14)	(0.23)	(0.21)	(0.47)	(0.47)	(0.24)	(0.21)	(0.27)	(0.25)	(0.28)	(0.26)
Population density, 1810		0.96***		1.27***		1.34***		1.42***		1.48***		1.51***
		(0.17)		(0.27)		(0.28)		(0.27)		(0.29)		(0.31)
R^2	0.27	0.38	0.23	0.31	0.12	0.15	0.22	0.31	0.22	0.30	0.19	0.26
Panel B. 2012–2021												
Carbonate-rich bedrock	0.31*	0.12	0.84***	0.58***	1.15**	0.88*	0.67***	0.38*	0.86***	0.56**	0.75**	0.44
	(0.16)	(0.14)	(0.24)	(0.21)	(0.48)	(0.49)	(0.25)	(0.22)	(0.28)	(0.26)	(0.30)	(0.27)
Population density, 1810		0.94***		1.30***		1.36***		1.46***		1.54***		1.56***
		(0.17)		(0.27)		(0.28)		(0.27)		(0.30)		(0.31)
R^2	0.28	0.38	0.24	0.32	0.11	0.15	0.21	0.31	0.22	0.30	0.18	0.26
Panel C. 2021												
Carbonate-rich bedrock	0.34**	0.16	0.95***	0.69***	0.86***	0.58***	0.80***	0.52**	1.03***	0.73***	0.84***	0.54*
	(0.16)	(0.14)	(0.24)	(0.22)	(0.24)	(0.22)	(0.25)	(0.23)	(0.28)	(0.26)	(0.30)	(0.28)
Population density, 1810		0.93***		1.30***		1.42***		1.44***		1.50***		1.53***
		(0.17)		(0.27)		(0.27)		(0.27)		(0.30)		(0.32)
R^2	0.29	0.39	0.25	0.32	0.23	0.32	0.21	0.30	0.23	0.30	0.19	0.26
Geographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	381	381	381	381	381	381	381	381	381	381	381	381

Notes: Restricted-access private-sector workplace records from Statistics Sweden. All outcomes are measured per square kilometer. Annual densities are averaged within each multiyear period before transformation. Within each outcome pair, the second column adds population density in 1810. Each regression uses one observation per historical parish or city; the geographic-control and observation rows apply to all panels. Carbonate-rich bedrock is a zero-to-one area share. Geographic controls comprise a full cubic polynomial in coordinates and a separate control for area. Standard errors in parentheses are heteroskedasticity-robust. Stars denote * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C Geographical Sensitivity

Table A3: Modern residential density and historical county fixed effects

	(1)	(2)	(3)	(4)
Panel A. Dependent variable: population density (2025)				
Carbonate-rich bedrock share	0.46**	0.46**	0.24	0.25
	(0.22)	(0.23)	(0.21)	(0.21)
Population density, 1810			1.08***	1.08***
			(0.22)	(0.23)
R^2	0.28	0.28	0.34	0.34
Panel B. Dependent variable: employed-resident density (2024)				
Carbonate-rich bedrock share	0.47**	0.47**	0.26	0.26
	(0.22)	(0.22)	(0.20)	(0.20)
Population density, 1810			1.06***	1.06***
			(0.22)	(0.22)
R^2	0.30	0.30	0.37	0.37
Location controls	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes
County fixed effects	No	Yes	No	Yes
Observations	381	381	381	381

Notes: Both panels use the same 381 historical areas. Employed residents are aged 20–64. Location controls are a full cubic coordinate polynomial. County fixed effects use the harmonized Malmöhus and Kristianstad classification described in Table 8. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

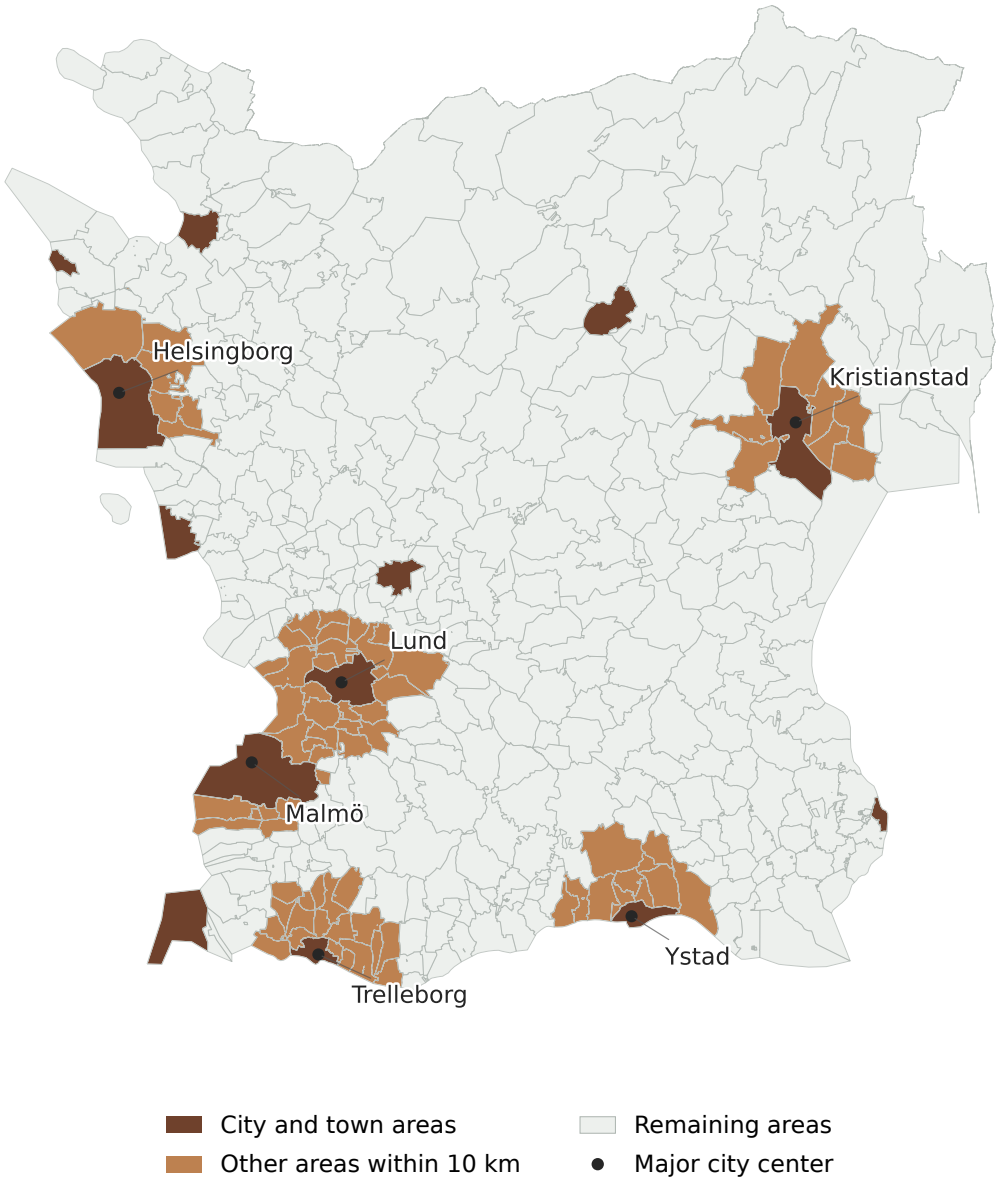


Figure A2: Cities, towns, and major-city neighborhoods

Notes: Brown areas comprise city units in the boundary layer, units classified as pre-1810 towns, and the six major-city parishes. Orange marks other areas whose representative interior point lies within 10 kilometers of a marked city center. Whole areas are shaded. Distances are straight-line distances in SWEREF 99 TM. Boundaries are from Lantmäteriet; town classifications use the historical-town layer.

Table A4: Spatial inference for historical population and agricultural capacity

Dependent variable	Coefficient	Parish-clustered standard error	Spatial standard errors (Conley) 25 km	50 km	100 km	Permutation p-value	Observations
Population density	0.18	0.06	0.09	0.09	0.06	< 0.01	1,905
Agricultural capacity (mantal density)	0.14	0.04	0.05	0.05	0.04	< 0.01	1,894

Notes: Coefficients are on carbonate-rich bedrock share in the pooled regressions of Table 2, with location and area controls and year fixed effects. Both regressions cover 381 historical areas. Spatial standard errors use distances between representative interior points and Bartlett weights with the indicated cutoffs, allowing correlation across areas and years. Two-sided permutation p-values compare the observed coefficients with 999 random reassignments of bedrock shares across areas. Each assigned share is held fixed across years; outcomes and controls are unchanged. The reassignments do not preserve spatial correlation in geology.

Table A5: Spatial inference for historical and modern density

Dependent variable	Coefficient	Heteroskedasticity-robust standard error	Spatial standard errors (Conley) 25 km	50 km	100 km	Observations
Panel A. Coefficient on carbonate-rich bedrock share						
Historical population density	0.18	0.03	0.09	0.09	0.06	1,905
Agricultural capacity (mantal density)	0.14	0.03	0.05	0.05	0.04	1,894
Population density, 1810	0.20	0.07	0.10	0.10	0.07	381
Panel B. Coefficient on population density in 1810						
Population density, 2025	1.08	0.22	0.23	0.22	0.20	381
Employed-resident density, 2024	1.06	0.22	0.22	0.21	0.19	381

Notes: All regressions include the baseline location and area controls. The first two rows of Panel A also include year fixed effects; Panel B controls for carbonate-rich bedrock share. Employed residents are aged 20–64. The heteroskedasticity-robust standard errors omit the finite-sample correction used in the main cross-sectional tables and do not allow within-area correlation in the pooled historical regressions; Table A4 reports the corresponding clustered standard errors. Spatial standard errors use Bartlett weights and distances between representative interior points, allowing correlation across areas and years. Every regression covers 381 historical areas; observation counts include repeated years where applicable.